

Baja.bio — Feature List

One workspace for genomics, oligo design and RNA prediction.

A browser-based genome editor for designing nucleic-acid therapeutics — visualize a gene, design the oligo, predict how it will behave, and check it against the literature and the patent landscape, all in one place.

Genome browser & visualization

- ✓ **Multi-track browser** — stacked, editable tracks with live parent/child sync
- ✓ **Genomics with mutations** — SNPs, indels and variants shown in genomic context
- ✓ **Load variants by source** — ClinVar, gnomAD, dbSNP, COSMIC over a selected region
- ✓ **Filter & manage SNPs** — by type, source and clinical significance (pathogenic, etc.)
- ✓ **Mutations navigator** — drill down by track / type / location and zoom to any variant
- ✓ **Create mRNA transcripts** — collapse exons into a spliced transcript

Oligo & assay design

- ✓ **siRNA & duplexes** — sense / antisense with 3' overhangs
- ✓ **ASO & gapmers** — chemistry-aware wing / DNA-gap / wing patterning
- ✓ **qPCR primers & probes** — designed with primer3, ranked by predicted assay success
- ✓ **Off-target analysis** — Levenshtein search across the transcriptome, both strands
- ✓ **Design-by-rules** — add and tile oligos across a target with live feedback
- ✓ **RNA secondary structure** — fold and design on stem-loops

Data & analysis layers

- ✓ **RNA-seq & IP layers** — coverage wiggles and peak / binding tracks
- ✓ **Protein & ORFs** — translate, find ORFs and work in protein space
- ✓ **Sequence tools** — edit sequence, annotate exons / mutations / sites, label & highlight regions
- ✓ **Patent & IP mapping** — overlay patent claims onto the sequence
- ✓ **Extract from a document** — upload a PDF / text / image and auto-load its genes and mutations

Library, chemistry & workspace

- ✓ **Reference bookshelf** — searchable RNA-therapeutics references and technical notes
- ✓ **Chemistry / compound editor** — nucleic-acid chemistries, conjugates and delivery
- ✓ **“The Chemistry of RNA Therapeutics”** — included with your workspace
- ✓ **Cloud workspace** — your files, tracks and designs, saved and synced
- ✓ **Guided feature tour** — see every capability at a glance

BAJA PREDICTION MODELS

Sequence-trained models, built into the browser

BajaSplice

Human splicing, four models.

- ✓ Splice sites at 0.935 PR-AUC
- ✓ Exon inclusion / PSI at 0.965 AUC
- ✓ Competing-site usage
- ✓ Cryptic-exon discovery (e.g. STMN2)
- ✓ 96% of ClinVar pathogenic splice variants flagged

BajaCLIP

RBP footprints from sequence.

- ✓ 170 RNA-binding proteins per 64-nt window
- ✓ 48 reliable RBPs at AUROC $\geq$ 0.90
- ✓ Per-position binding profile for any RBP
- ✓ Trained on ENCODE eCLIP + POSTAR3
- ✓ No genome or download required

djPrimer

qPCR assay-success prioritization.

- ✓ Design with primer3, prioritize with this
- ✓ Expression-aware success prediction
- ✓ Ranks which designed assays will validate
- ✓ Measured on 2,800+ real primer sets
- ✓ Roughly halves failed validations

Each model ships with the app and renders directly onto the browser's tracks and sashimi layers. Technical notes for all three are in the reference library.